

Multiple Choice

1. **Answer:** C

Options: A) 1; B) 2; C) 2,3; D) 6

Note: The correct answer is 2 and 3 because substituting these values into the polynomial $x^3 + 2x + 2$ in the finite field $\mathbb{Z}_7$ results in zero, confirming that they are indeed the roots of the polynomial.

2. **Answer:** D

Options: A) 1; B) 2; C) 3; D) 4

Note: According to Fermat's Little Theorem, since 23 is a prime and does not divide 3, we have that $3^{22} \equiv 1 \pmod{23}$, allowing us to reduce 3^{47} to $3^{(47 \bmod 22)} = 3^3$, which equals 27, and the remainder when 27 is divided by 23 is 4.

3. **Answer:** B

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Both statements are false because a homomorphism can have a non-empty kernel (for example, the trivial homomorphism where every element maps to the identity) and it is indeed possible to have a nontrivial homomorphism from a finite group into an infinite group (for instance, a finite cyclic group can map to an infinite cyclic group).

4. **Answer:** D

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Statement 1 is false because not every set of elements forms an extension field of $\mathbb{Z}_2$, while Statement 2 is true as it reflects the fundamental theorem of algebra, which asserts that every non-constant polynomial over a field has at least one root in some extension field.

5. **Answer:** B

Options: A) the entire real axis; B) a point; C) a ray; D) an open finite interval

Note: Since an analytic function that maps the entire finite complex plane into the real axis must be constant according to Liouville's theorem, it follows that the imaginary axis, being part of the domain, can only be mapped to a single point on the real axis.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: The order of the finite group must be a multiple of the order of the subgroup (5) and must also satisfy the condition of having an even number of elements, which is possible with 35, as it allows for elements whose orders meet the criteria without having non-identity elements that are their own inverses.

7. **Answer:** True

Options: A) True; B) False

Note: The polynomial $8x^3 + 6x^2 - 9x + 24$ satisfies the Eisenstein criterion for irreducibility with $p=3$ because the coefficients of the polynomial meet the conditions where all coefficients except the leading coefficient are divisible by 3, and the constant term is not divisible by 9.

8. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the degree of the sum of two polynomials is not necessarily equal to the sum of their degrees; specifically, if the leading coefficients of $f(x)$ and $g(x)$ cancel each other out, the degree of $f(x) + g(x)$ can be less than that of either $f(x)$ or $g(x)$.

9. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the degree of the field extension $\mathbb{Q}(2, 3)$ over $\mathbb{Q}$ is actually 4, as the extensions $\mathbb{Q}(2)$ and $\mathbb{Q}(3)$ are each of degree 2 over $\mathbb{Q}$, and they are linearly independent over $\mathbb{Q}$.

10. **Answer:** False

Options: A) True; B) False

Note: The statement is false because there exist rings, known as "rings without identity," that do not include a multiplicative identity element.

Long Form Response

11. **Answer:** The statement regarding subgroups and group order is valid, particularly in the context of the formula $|HK| = |H||K|/|H \cap K|$, which describes the order of the product of two subgroups H and K within a group G . This formula highlights how the intersection of H and K affects the overall size of the subgroup generated by their product, emphasizing that the order of the combined subgroup is influenced by the overlap of elements in H and K . Moreover, groups of order $2p$, where p is a prime, exhibit particular structures, often leading to the existence of normal subgroups and facilitating the application of Sylow's theorems. These characteristics reinforce the validity of the statements, illustrating the intricate relationship between subgroup orders and group structure.

Note: The answer correctly identifies that the formula $|HK| = |H||K|/|H \cap K|$ illustrates how the sizes of two subgroups H and K and their intersection determine the order of the subgroup they generate, and it also begins to address the structural characteristics of groups of order $2p$, linking subgroup interactions to group theory.

12. **Answer:** The characteristic polynomial of the matrix A is given by $\det(A - I) = -\lambda^3 + 3\lambda^2 + -3$. From this polynomial, we can extract important information about the matrix: the trace of A , which is the sum of its eigenvalues, is equal to the coefficient of λ^2 (with the opposite sign), the

3. The constant term, which is the determinant of A , is -3 , indicating that A is not invertible. To determine the eigenvalues of A , we solve the characteristic equation $\det(A - \lambda I) = 0$. Note: correct because it accurately identifies the relationship between the coefficients of the characteristic polynomial and the eigenvalues. The characteristic polynomial is $\lambda^3 - 3\lambda^2 + 3\lambda - 3 = 0$, which together provide insights into the properties and eigenvalues of the matrix.

End of Answer Key